

Data Engineering: Deep Learning

Responsible AI: Legal & Ethical Challenges

Master's in Information Engineering

ISEP



Setting the Scene

What are the legal & ethical challenges of AI?



State Farm hit with lawsuit over AI bias, discrimination and unpaid insurance claims

 Nataly Kramer Oct 15, 2025 ⌚ 2 min

Fonte: Beinsure

Airline held liable for its chatbot giving passenger bad advice - what this means for travellers

23 February 2024

By Maria Yagoda, Features correspondent

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Fonte: BBC

CODED BIAS



Amazon ditched AI recruiting tool that favored men for technical jobs

Specialists had been building computer programs since 2014 to review résumés in an effort to automate the search process

Fonte: The Guardian

MIT
Technology
Review

Artificial Intelligence / Machine Learning

Biased Algorithms Are Everywhere, and No One Seems to Care

The big companies developing them show no interest in fixing the problem.

by Will Knight

Jul 12, 2017

Image & Art as
input for AI
training

AI Firms Face Growing List of Lawsuits. Here's What to Watch

- Elon Musk is latest to sue OpenAI, claiming CEO to
- Journalists have accused the company of copyright

By [Antonia Mufarech](#) and [Ethan M Steinberg](#)

March 1, 2024 at 10:09 PM GMT

Source: Bloomberg

INTELIGÊNCIA ARTIFICIAL

Artistas portugueses analisam formas de abordar uso indevido de obras por IA

Regina Pessoa, Jorge Jacinto e Vhils são alguns dos nomes que integram a lista de mais de 4700 artistas cujos trabalhos foram utilizados, sem autorização, para treinar a plataforma Midjourney.

Source: Público

“Art” of AI

The Legal Challenges of Generative

AI deepfakes increasing fraud risks for businesses, KPMG survey finds

As fraudsters use emerging technologies to exploit vulnerabilities, the need to buttress fraud defences mo

Source: KPMG

Deep Fakes

Prompt: A robot painting a painting of a robot

How to stop AI deepfakes from sinking society – and science

Deceptive videos and images created using generative AI could sway elections, crash stock markets and ruin reputations. Researchers are developing methods to limit their harm.

Source: Nature

Other risks & Challenges of AI

Beyond bias, AI introduces a wide spectrum of risks that organisations must actively manage. Regulation alone cannot address all of them.



Societal

Deepfakes, misinformation, erosion of trust, and concentration of power



Tech

Hallucinations, model drift, data poisoning, and brittleness



Operational

Over-reliance on automation and lack of human oversight



Legal

IP infringement, GDPR conflicts, liability exposure



Strategy

Misalignment between AI adoption and business objectives



Cybersecurity

Adversarial attacks, prompt injection, and model theft

Compliance vs Ethical Use of AI



Ethical framework

Advantages

Flexible and enforced through reputation

Disadvantages

- Partially applicable due to lack of legitimacy
- Risk of buying ethics and whitewashing practices

Exempls:

- AIHLEG ALTAI
- OECD – G20
- UNESCO
- IEEE CertifAIEd framework

Legal framework

- Mandatory
- Source of legal and regulatory obligations



Have you heard about the AI
Act?

MEPs approve world's first comprehensive AI law

4 days ago

By Shiona McCallum, Liv McMahon & Tom Singleton,

riters

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Europe's landmark AI Act passes Parliament vote

By Eliza Gkritsi | Euractiv ⌚ Est. 2min

📅 Mar 13, 2024

Content-Type: News



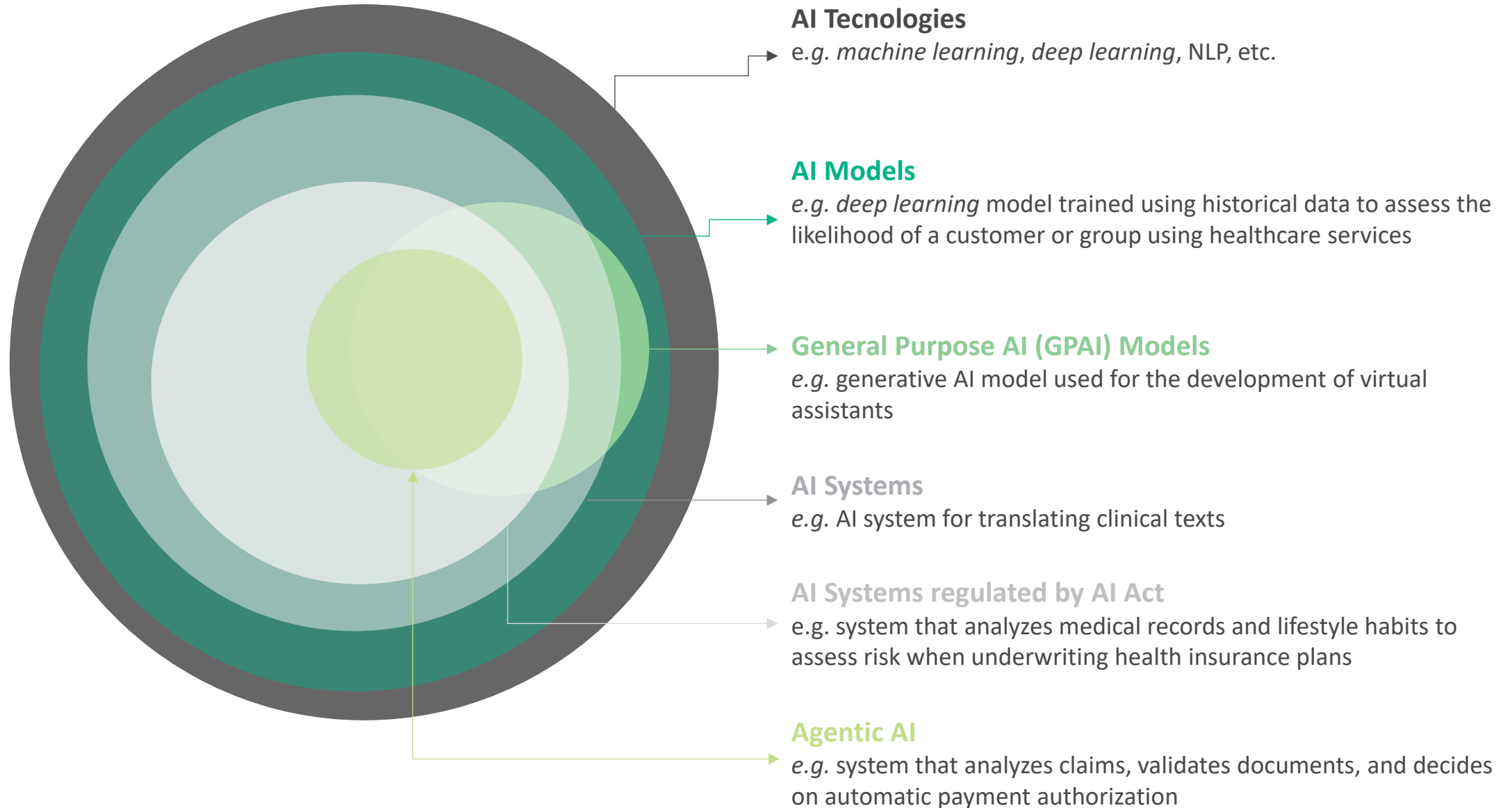
Co-rapporteur Dragoș Tudorache after the vote on the AI Act at the E.U. Parliament on March 13.

Source: Euractiv



Source: BBC

AI Technology, Models, Systems, Agents





AI Act



BROAD SCOPE OF APPLICATION

Horizontal application with obligations for most **stakeholders** in the **AI value chain** and **extraterritorial** scope of application covering AI systems outside the EU



RISK-BASED APPROACH

Prohibited risk » High-risk » Limited / Specific Risk (including a tiered-approach for GPAI & GenAI) » Minimal Risk



NEW OBLIGATIONS

Providers and **deployers** of high-risk systems are particularly impacted by the new rules and are subject to obligations such as risk assessment, transparency, human supervision, training, data governance for training and testing datasets



NEW AUTHORITIES

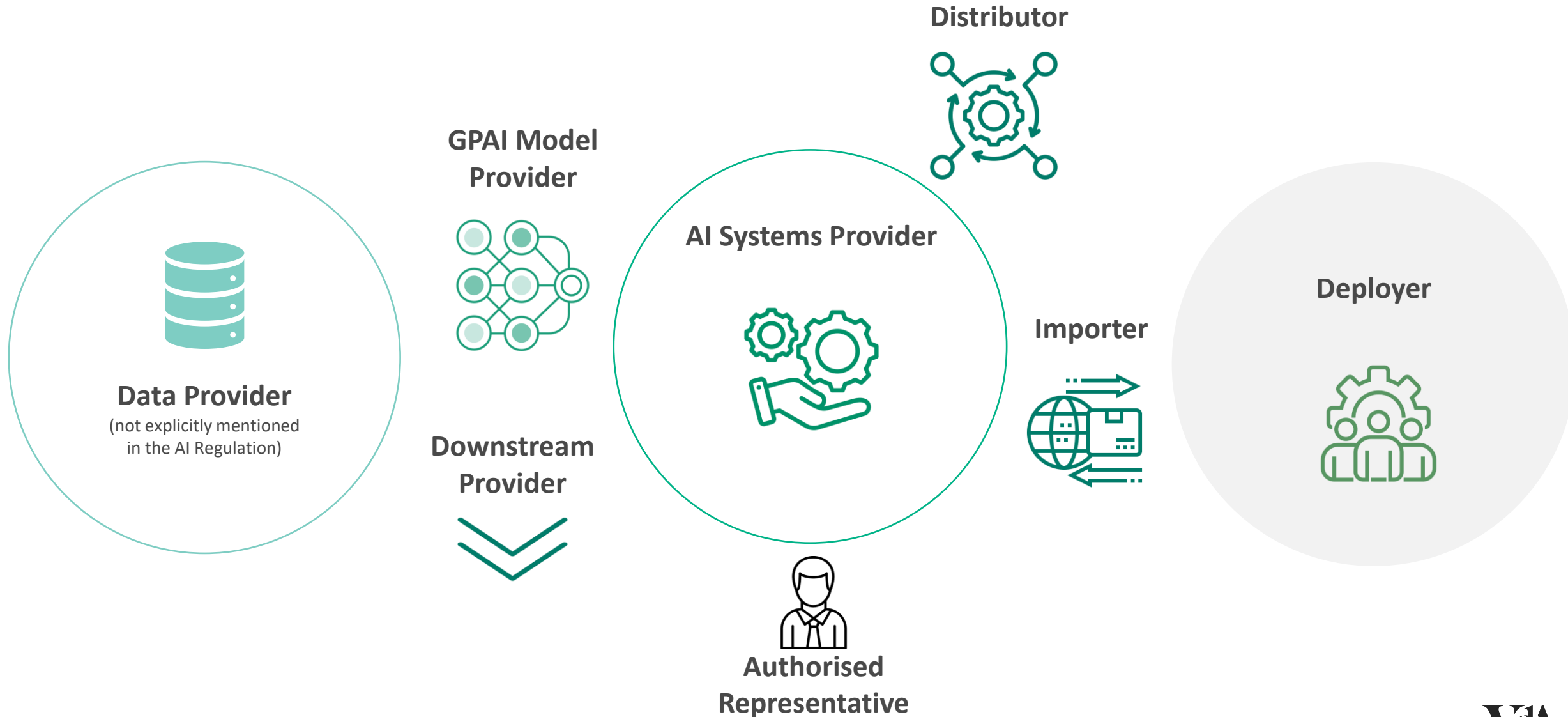
Every **MS** should appoint a new national authority responsible for supervising the application and implementation of the of the Regulation (**ANACOM**). New EU authorities are also created: **EU AI Office & EU AI Board**



FINES

35 000 000 EUR or until **7%** of the total annual worldwide turnover for the previous financial year, whichever is the higher.

AI Ecosystem: The Value Chain Under the AI Regulation



AI Act: Objectives

Risks to people

Health



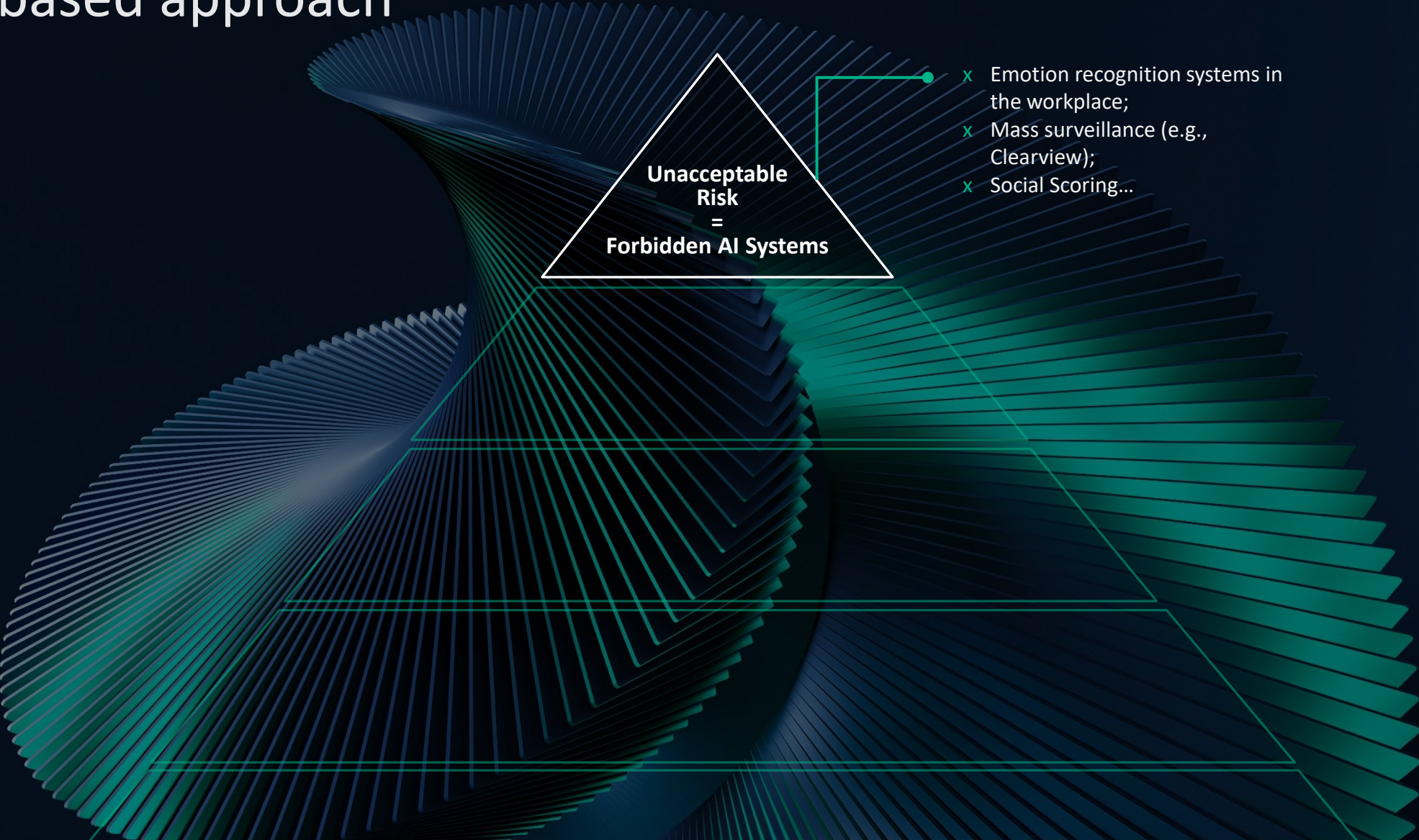
Safety



**Fundamental
Rights**



Risk-based approach



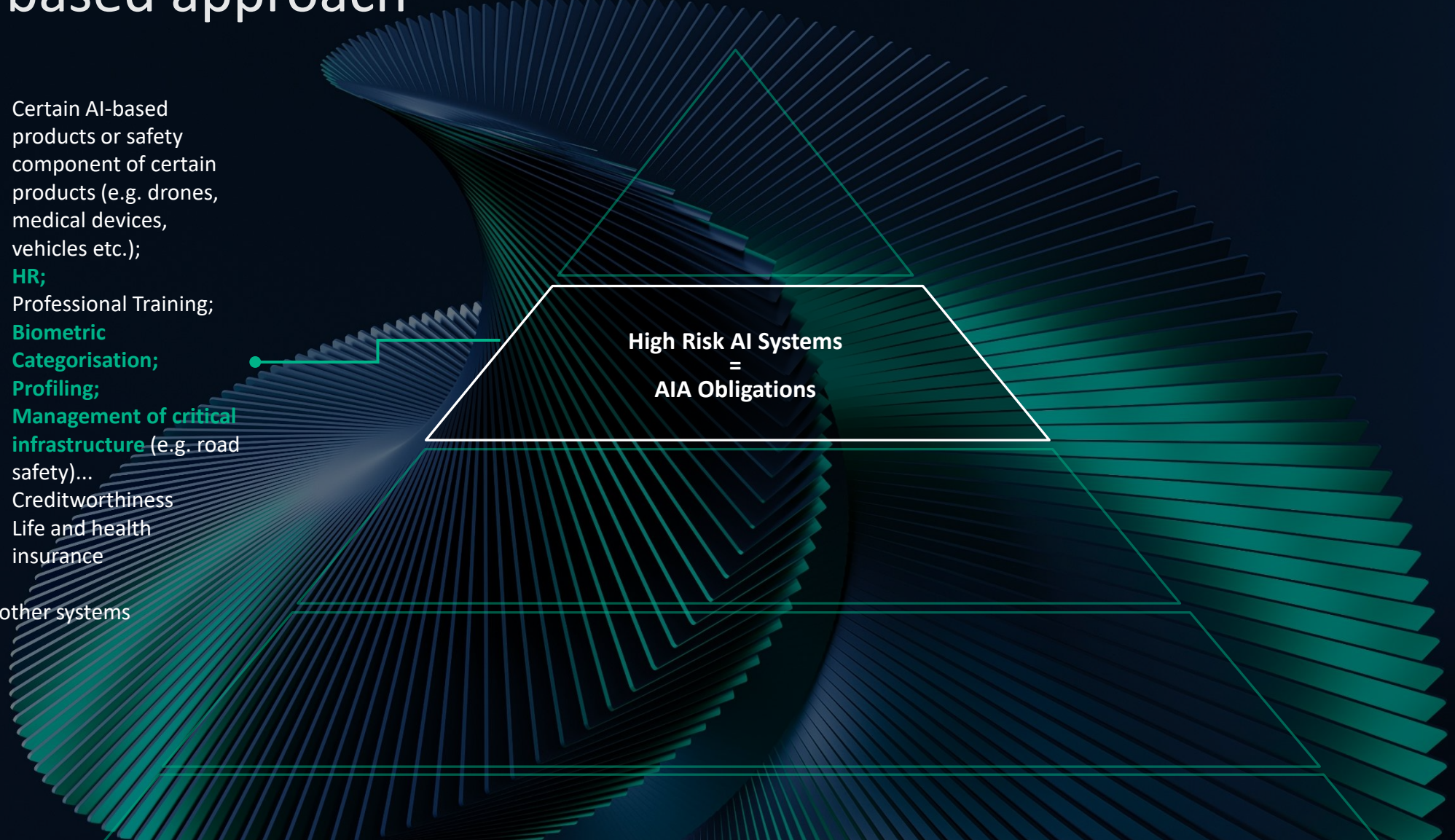
Unacceptable
Risk
=
Forbidden AI Systems

- x Emotion recognition systems in the workplace;
- x Mass surveillance (e.g., Clearview);
- x Social Scoring...

Risk-based approach

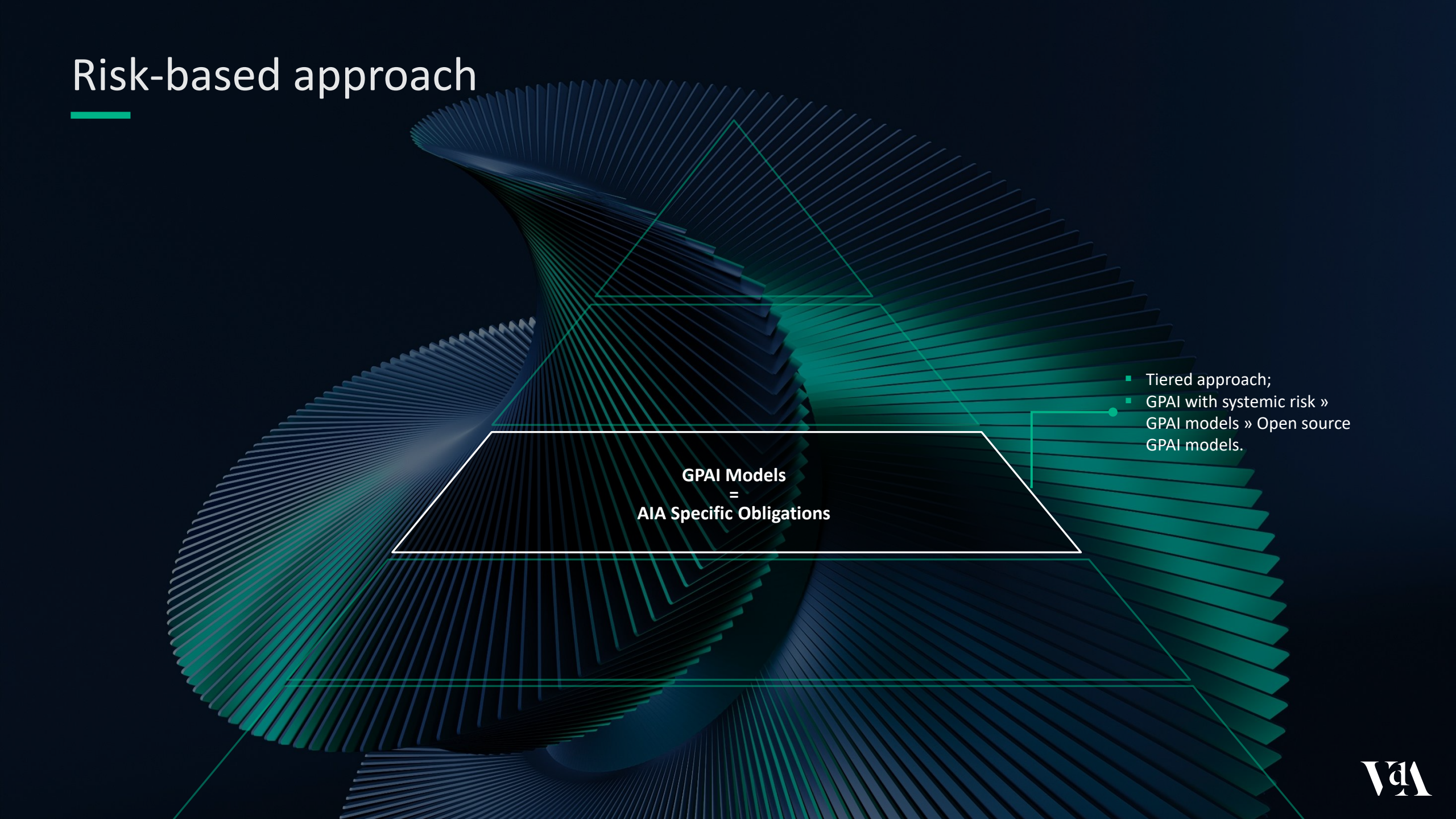
- Certain AI-based products or safety component of certain products (e.g. drones, medical devices, vehicles etc.);
- **HR;**
- Professional Training;
- **Biometric Categorisation;**
- **Profiling;**
- **Management of critical infrastructure** (e.g. road safety)...
- Creditworthiness
- Life and health insurance

+ other systems



High Risk AI Systems
=
AIA Obligations

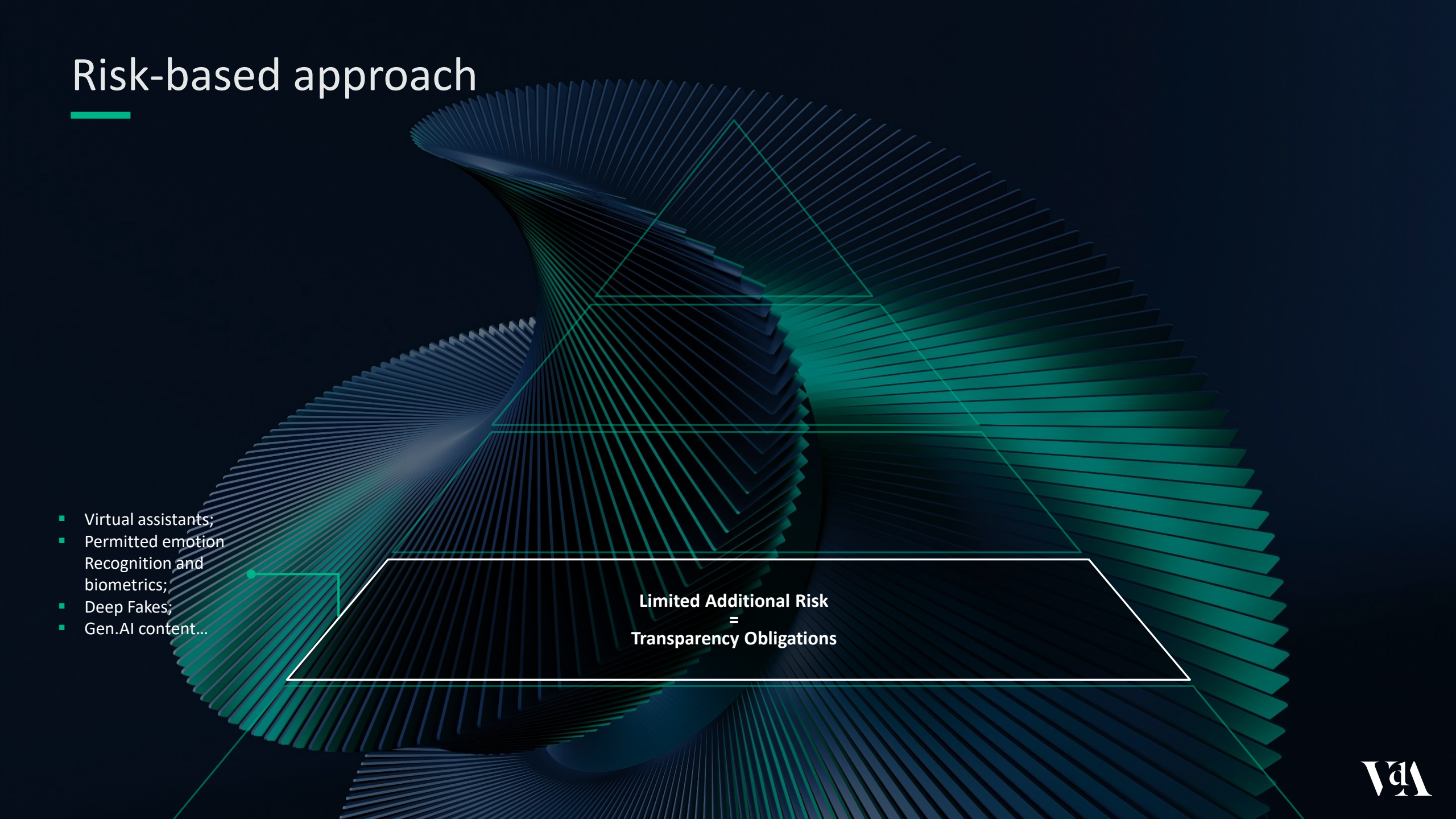
Risk-based approach



GPAI Models
=
AIA Specific Obligations

- Tiered approach;
- GPAI with systemic risk » GPAI models » Open source GPAI models.

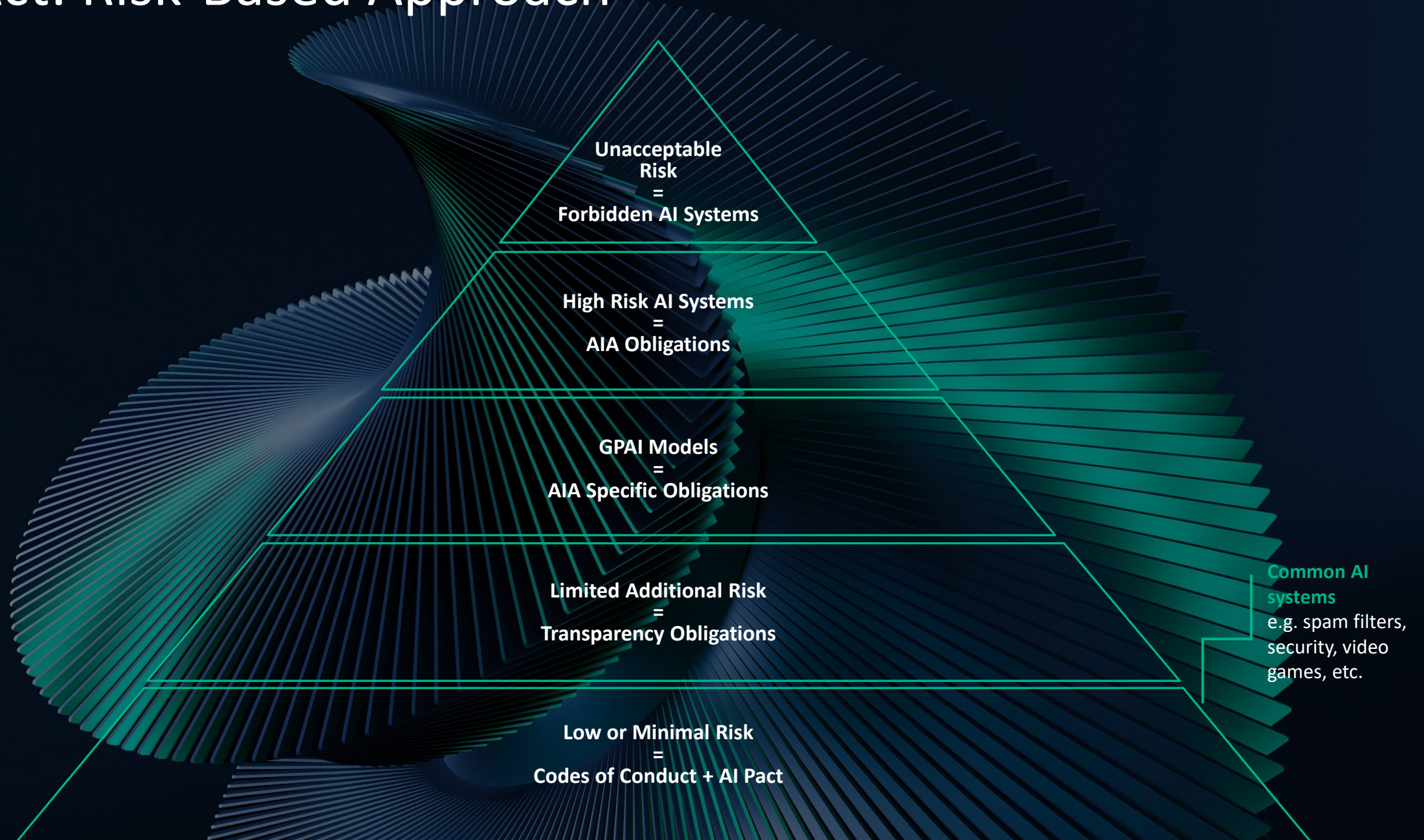
Risk-based approach



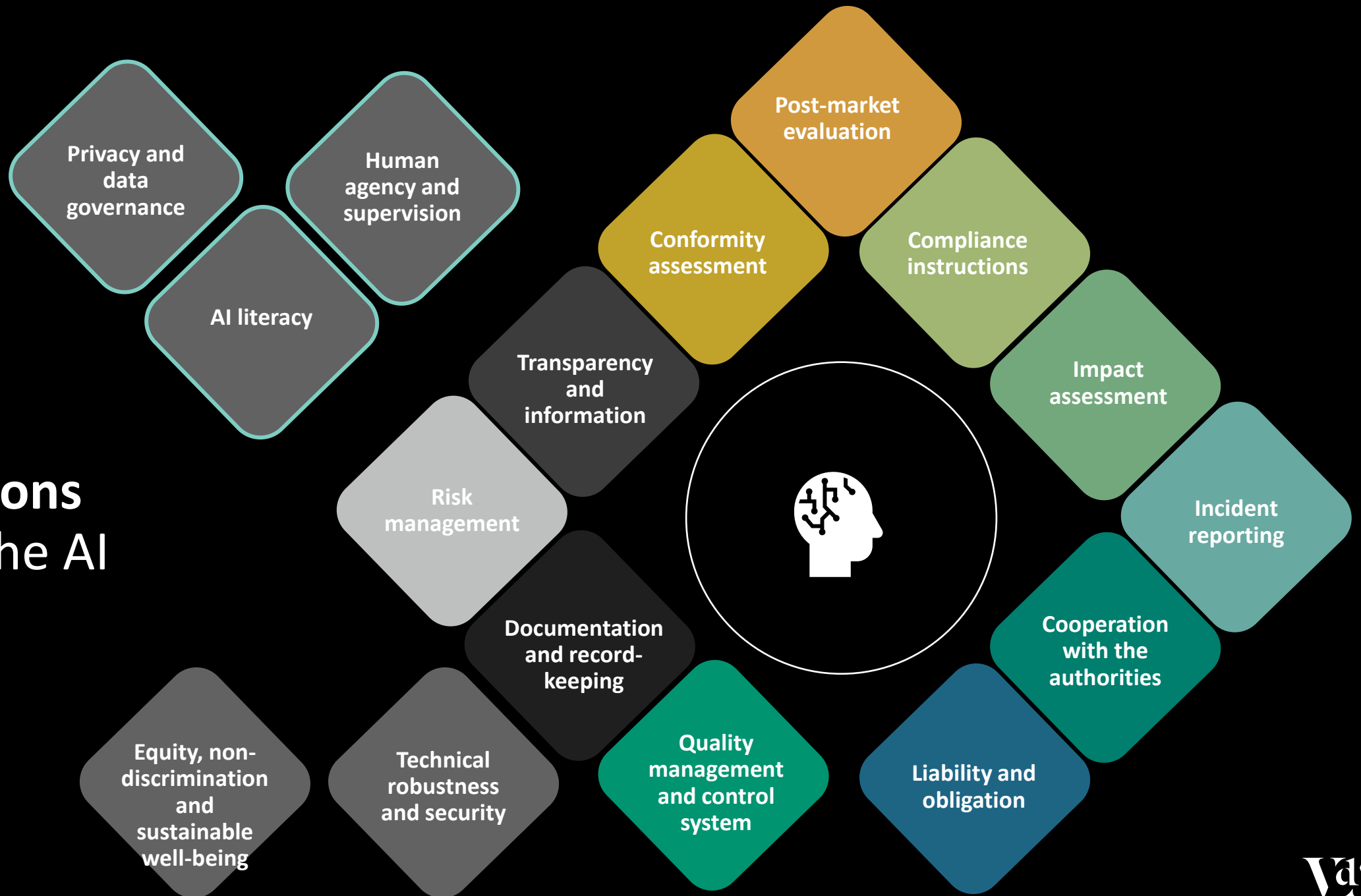
- Virtual assistants;
- Permitted emotion Recognition and biometrics;
- Deep Fakes;
- Gen.AI content...

Limited Additional Risk
=
Transparency Obligations

AI Act: Risk-Based Approach



Some obligations under the AI Act



Application Phases

Effective date 1 August 2024



AI Omnibus!

How to build Responsible AI?

Building Responsible AI: Practical Principles

Whether or not your system is classified as high-risk, embedding these principles from the earliest prototype stage will make your solution more robust, trustworthy, and deployable at scale.



Design for Explainability from Day One

Choose model architectures that allow you to explain outputs — not just optimise accuracy metrics



Audit Your Training Data

Actively test datasets for representation gaps, historical bias, provenance, copyright, and distribution shift before training begins



Build Human Oversight In, Not On

Design override mechanisms, confidence thresholds, and escalation paths into the product architecture — not as afterthoughts



Document Everything

Maintain model cards, data sheets, risk registers, and decision logs from your first sprint - they will be required at scale



Monitor Post-Deployment

Deploy with feedback mechanisms, drift detection, and incident reporting built into your MLOps pipeline



A strong IP strategy is as important as your compliance strategy. Registering your trademarks, filing patents early, and documenting your creative process are not just legal formalities — they are the foundation of your competitive moat. Startups that treat IP as an afterthought often find themselves unable to defend their innovations or attract serious investors.

Q&A



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